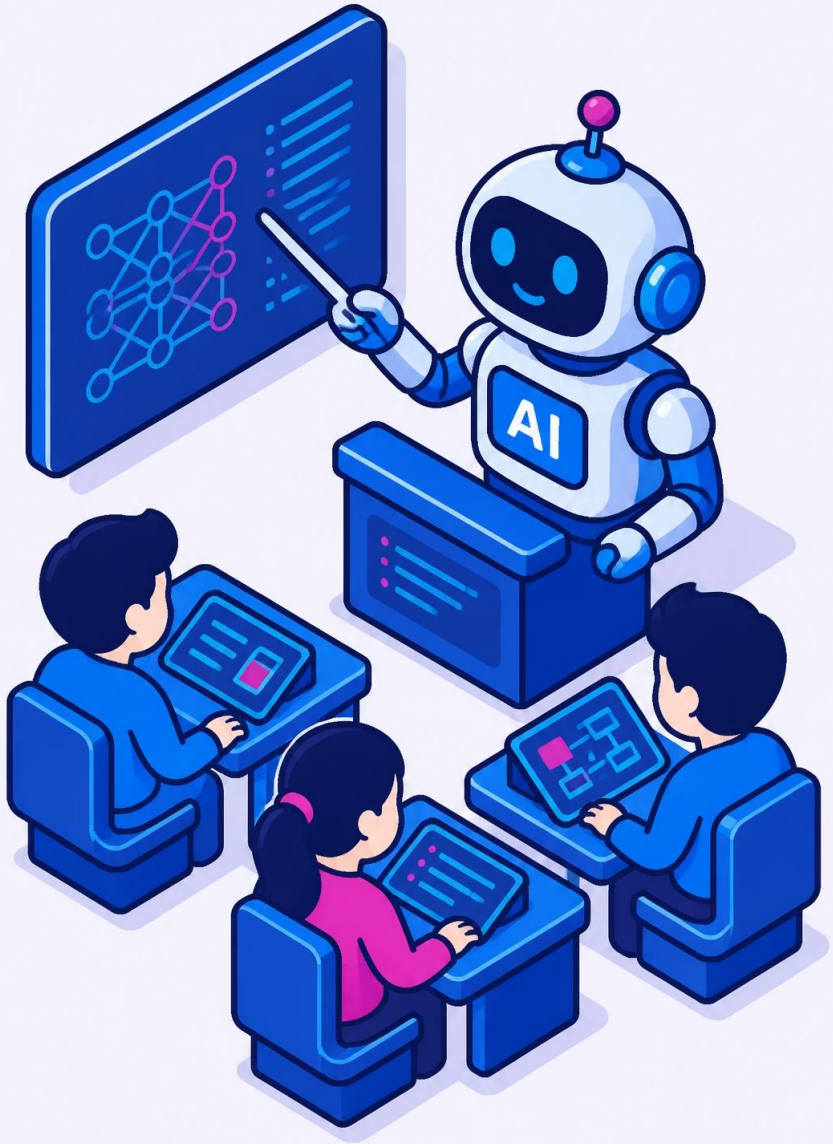


EvalTutor – AI-Powered Multilingual Tutoring Evaluation Platform

B.Tech Final Year Project Presentation | Academic Year 2025–2026

Student Details

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Institutional Details

- **College Name:** [ANNA UNIVERSITY REGIONAL CAMPUS]
- **Guide:** [Dr.Marikannu]
- **Academic Year:** 2025–2026

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The Problem: AI Tutors Are Unmonitored & Unreliable

AI tutors now serve millions of students globally — yet there is **zero pedagogical oversight** on the content they generate. The consequences are serious and growing.

🚀 Explosive AI Adoption

AI tutors serve millions of students globally, yet response quality remains entirely unchecked and unmonitored.

🧠 Hallucinations Are Rampant

LLMs confidently generate incorrect facts, misleading students during critical learning phases with no warning.

⚠️ Dangerous Dependency

Students copy AI answers without understanding — directly undermining conceptual learning and critical thinking.

🌐 Language Barrier

Most AI tutors operate only in English, excluding **1.5 billion+** regional-language learners worldwide.

🔍 No Evaluation Layer

There is currently **zero** pedagogical oversight or quality scoring on AI-generated tutoring content.

✅ EvalTutor Fills the Gap

A real-time evaluation platform that monitors, scores, and continuously improves AI tutoring quality.

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Objectives

EvalTutor is designed around five core objectives that together create a comprehensive, trustworthy AI tutoring ecosystem.



Monitor AI Quality

Score every AI response for pedagogical accuracy, clarity, and educational value on a 0–100 scale.



Detect Hallucinations

Flag responses with high hallucination risk before students act on potentially incorrect information.



Measure Dependency

Track over-reliance on AI answers versus independent reasoning to promote genuine learning.



Multilingual Support

Enable seamless tutoring in regional languages via automatic language detection and translation.



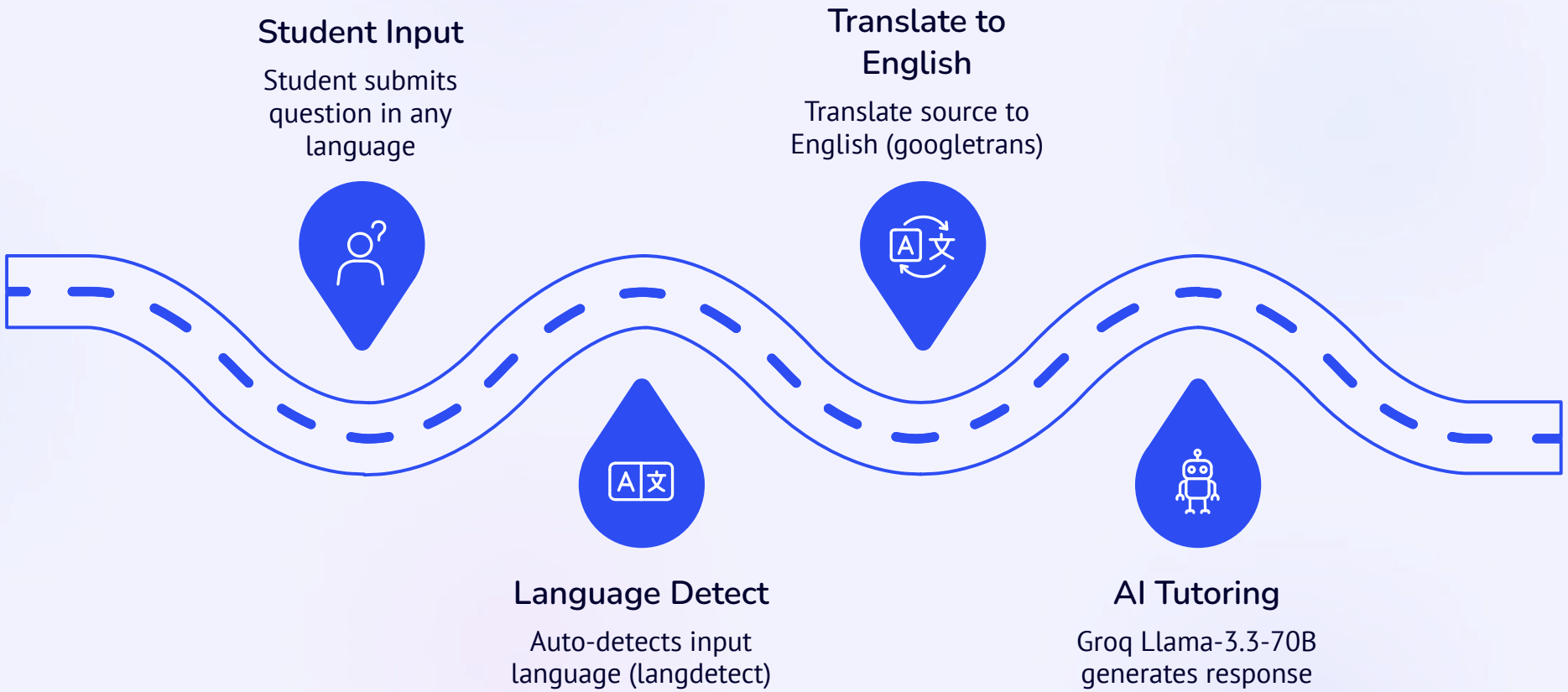
Analytics Dashboard

Provide students and educators with real-time learning insights, progress metrics, and weak topic identification.

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How EvalTutor Works: End-to-End Workflow

A seamless 7-step pipeline transforms a student's question — in any language — into an evaluated, scored, and translated AI tutoring response delivered in real time.



Each step is automated and executes in milliseconds — the student experiences a seamless, real-time tutoring interaction while the evaluation engine works silently in the background to score every response.

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System Architecture

EvalTutor's architecture is modular and layered — each component is independently scalable and communicates through a **FastAPI** orchestration backbone.



Core Processing Modules

- **Langdetect:** Identifies input language in milliseconds
- **Googletrans:** Bridges regional languages ↔️ English
- **Groq Llama-3.3-70B:** Generates pedagogically structured responses

Evaluation & Output Layer

- **Evaluation Engine:** Scores quality, detects hallucinations, measures dependency
- **Dashboard:** Chart.js-powered real-time analytics
- **FastAPI:** Handles all routing and session management

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Technologies Used

EvalTutor is built on a modern, lightweight, and high-performance technology stack — each layer chosen for speed, accuracy, and developer productivity.

Layer	Technology
Frontend	HTML, CSS, JavaScript
Backend	FastAPI, Python
AI Model	Groq Llama-3.3-70B-Versatile
Translation	Googletrans, Langdetect
Charts	Chart.js
Prompt Engineering	Structured JSON & Markdown



Ultra-Fast Inference

Groq's hardware-accelerated API delivers AI responses in milliseconds — enabling truly real-time tutoring interactions.



Python-First Stack

FastAPI and Python provide a clean, async-ready backend that integrates seamlessly with all AI and translation modules.



Live Visualization

Chart.js renders dynamic, real-time analytics charts on the student dashboard after every single interaction.

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Key Features

Seven tightly integrated features make EvalTutor a complete, end-to-end AI tutoring quality platform — not just a chatbot wrapper.

⚡ Live AI Chat

Real-time tutoring powered by Groq's ultra-fast inference API with structured educational responses.

🌐 Multilingual Support

Auto-detects and responds in the student's native language — inclusive for 1.5B+ regional learners.

📊 Pedagogical Evaluation

Every response scored against educational quality standards on a 0–100 scale in real time.

🚨 Hallucination Detection

Flags factually risky AI responses (Low / Medium / High) before they mislead students.

🔗 Dependency Tracking

Monitors how often students rely on AI versus thinking independently — promoting genuine learning.

📈 Student Dashboard

Personalized view of quality scores, weak topics, and learning progress over time.

🔄 Real-Time Analytics

Live charts and metrics updated after every single interaction using Chart.js visualizations.

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Working Methodology: 10-Step Process

From question submission to dashboard update — EvalTutor's complete processing pipeline executes automatically in under two seconds per interaction.

01

Student Submits Question

Student types a question in any language via the chat interface.

02

Language Identification

Langdetect identifies the input language automatically.

03

Query Translation

Googletrans translates the query to English for AI processing.

04

AI Response Generation

Groq Llama-3.3 generates a structured, pedagogically sound AI response.

05

Response Translation Back

Response is translated back to the student's native language.

06

Evaluation Engine Analysis

Evaluation Engine analyzes response quality across multiple dimensions.

07

Quality Score Calculated

Quality Score is computed on a 0–100 scale using a multi-dimensional rubric.

08

Hallucination Risk Flagged

Hallucination Risk is classified as Low, Medium, or High per response.

09

Dependency Level Logged

Dependency Level is measured and recorded for the session.

10

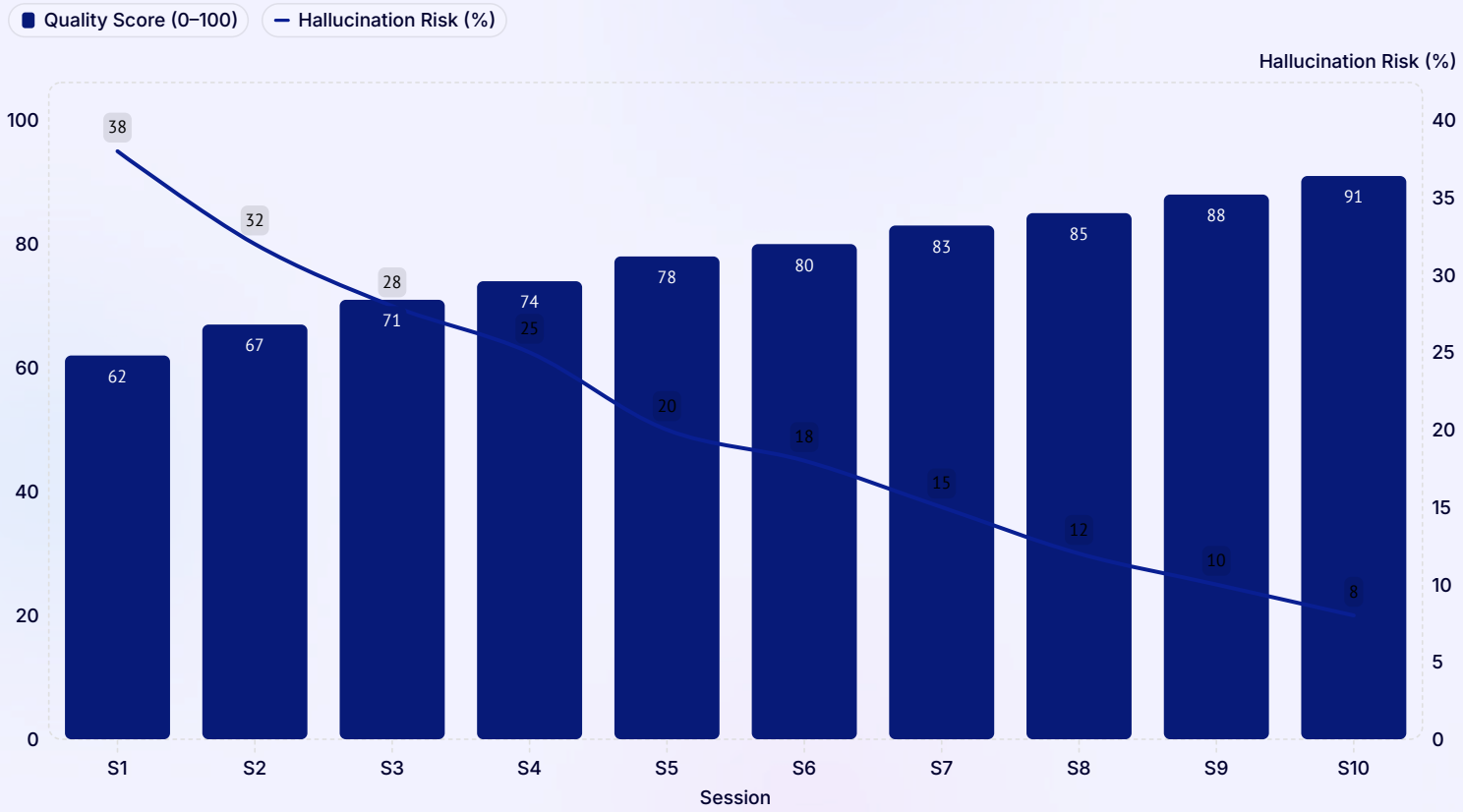
Dashboard Updated

All metrics are pushed to the student dashboard in real time via Chart.js.

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Results & Dashboard Insights

Across 10 sessions, EvalTutor demonstrates a clear and consistent improvement in AI tutoring quality — with **Quality Scores rising from 62 → 91** and **Hallucination Risk dropping from 38% → 8%** as students engage more deeply with the platform.



91

Peak Quality Score

Achieved by Session 10 on the 0–100 pedagogical accuracy scale.

8%

Hallucination Risk

Reduced from 38% at Session 1 — a **79% reduction** in risk.

3

Evaluation Dimensions

Quality Score, Hallucination Risk, and Dependency Level tracked per response.

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Challenges Faced & Solutions Implemented

Four significant engineering challenges were encountered and resolved during development — each solution strengthening the platform's robustness and accuracy.

Challenge	Solution Implemented
Groq model deprecation mid-development	Migrated to **Llama-3.3-70B-Versatile** — faster and more accurate than the deprecated model
Translation accuracy for technical terms	**English-first middleware layer** with context-aware prompts to preserve technical terminology
Poor AI response formatting	**Prompt engineering** with structured Markdown and JSON output templates for consistent formatting
Measuring educational quality objectively	**AI-based evaluation engine** with a multi-dimensional scoring rubric covering accuracy, clarity, and depth

Each challenge encountered during development directly informed a stronger, more production-ready architecture — making EvalTutor more resilient and accurate than originally designed.

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Advantages & Future Scope

✅ Current Advantages

- Improves AI tutoring quality with **real-time evaluation** after every response
- Detects and flags hallucinations **before** they mislead students
- Encourages conceptual learning over answer copying
- Reduces unhealthy student dependency on AI
- Supports regional languages — **inclusive for all learners**
- Provides actionable learning analytics for educators

🚀 Future Scope

- **Voice-Based Tutoring** — Speech-to-text integration for hands-free learning
- **OCR Support** — Handwritten question input via optical character recognition
- **Mobile Application** — Native iOS & Android apps for on-the-go learning
- **LMS Integration** — Moodle and Google Classroom compatibility
- **Personalized AI Paths** — Adaptive learning journeys per student profile
- **Offline Multilingual Models** — Low-connectivity region support

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